

STUDENT EDITION

Design Thinking in the Age of AI

How a human-centred problem-solving framework gets supercharged by Artificial Intelligence

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AGENDA

What we'll cover in 60 minutes

Part 1 • Foundations

- Ice-breaker activity
- What is Design Thinking?
- Why most solutions fail
- The 5 phases (deep dive)
- The iterative loop

Part 2 • AI as Partner

- What is AI? Plain English
- Generative AI explained
- AI at every phase
- Before vs After AI
- The human role

Part 3 • Practice & Future

- Real case studies
- Risks & responsible design
- Practical toolkit
- Your future career
- Key takeaways & Q&A

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Let's start with a quick question

"Think of a product or app you use every day. Why do you love it?"

Turn to the person next to you. Share your answer in 60 seconds each.

- What need does it meet for you?
- Did someone clearly understand your problem before building it?
- What would it look like if it had been built *without* understanding your needs?

WHY THIS MATTERS

The products we love exist because someone took the time to deeply understand the person using them. That is Design Thinking in action.

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PART 1 · FOUNDATIONS

What is Design Thinking?

"Design Thinking is a human-centred approach to innovation — it puts the needs of real people at the heart of every decision."

— Tim Brown, IDEO

- A **structured way of solving problems** used by the world's leading companies
- Born at **Stanford d.school** in the 1980s, popularised by design firm **IDEO**
- Used at Apple, Google, Airbnb, IBM, and even hospitals and governments
- **Not just for designers** — engineers, product managers, doctors, and educators use it
- Core belief: *understand people first, build solutions second*

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Why do most solutions fail?

42%

of startups fail because there's no real market need for what they built

70%

of digital transformation projects fail due to poor understanding of user needs

\$2.26T

wasted annually on poorly designed software in the US alone

ROOT CAUSE

Most failures are not engineering failures. They are *empathy failures* — someone built a solution without truly understanding the problem or the people who have it.

The 5 Phases of Design Thinking

A flexible, iterative process — not a rigid checklist. You can (and should) loop back at any point.

1

Empathize

Understand the real experiences and needs of your users

2

Define

Clearly frame the problem you are actually solving

3

Ideate

Generate many creative solutions without judgment

4

Prototype

Build quick, cheap versions to explore your best ideas

5

Test

Get real feedback from users and improve your solution

KEY INSIGHT

The "double diamond" — first *expand* your understanding (empathize → define), then *expand* ideas and *narrow* to a solution (ideate → test).

PHASE 1

Empathize — Walk in their shoes

Goal: Understand users so deeply you can see the world through their eyes.

Methods

- In-depth interviews (not surveys!)
- Observation in natural settings
- Immersion — experience it yourself
- Empathy mapping (thoughts, feelings, actions)
- Shadowing and contextual inquiry

Outputs

- User personas
- Empathy maps
- Journey maps
- Insights & themes
- Quotes and stories from users

COMMON MISTAKE

Asking users what they *want* instead of observing what they *do*. People are often poor predictors of their own behaviour.

PHASE 2

Define — Frame the right problem

Goal: Synthesise your research into a clear, meaningful problem statement.

"If I had an hour to solve a problem, I'd spend 55 minutes thinking about the problem and 5 minutes thinking about solutions."

— Attributed to Albert Einstein

- ▶ **Point of View (POV) statement:** [User] needs [need] because [insight]
- ▶ **"How Might We" questions** turn problems into opportunities
- ▶ A well-defined problem is half-solved
- ▶ Avoid jumping to solutions before the problem is clear

EXAMPLE

Bad: "Build a better bus app." Good: "Commuters in our city need to feel confident about their journey because unexpected delays cause them significant stress."

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PHASE 3

Ideate — Quantity over quality (at first)

Goal: Generate as many ideas as possible before evaluating any of them.

Rules of Brainstorming

- Defer judgment — no bad ideas yet
- Go for volume — aim for 100 ideas
- Encourage wild ideas
- Build on others' ideas
- Stay focused on the problem

Techniques

- Classic brainstorming
- Crazy 8s — 8 ideas in 8 minutes
- SCAMPER method
- Worst Possible Idea (reverse thinking)
- Mind mapping

KEY INSIGHT

The best idea is rarely the first idea. Diverge widely first, then converge to the most promising concepts.

PHASE 4

Prototype — Make it real, fast

Goal: Build the cheapest, fastest version that lets you test your key assumptions.



Paper Prototype

Sketch screens on paper.
Fastest way to test flows
— no coding required.



Wireframes

Low-fidelity digital
layouts — focuses on
structure, not aesthetics.



Clickable Mockup

High-fidelity prototype in
tools like Figma — feels
like the real thing.



Physical Model

For hardware or service
design — cardboard,
foam, or even role-
playing.

GOLDEN RULE

Spend only as much time on a prototype as needed to answer the question you're testing. Don't polish what you might throw away.

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PHASE 5

Test — Learn from real people

Goal: Get honest feedback to refine, improve, or sometimes restart.

- **Show, don't explain** — let users interact with the prototype, don't guide them
- Ask "**what are you thinking?**" as they use it — think-aloud protocol
- Watch what they *do*, not just what they say
- **5 users** reveal ~85% of usability issues (Nielsen's Law)
- Document observations, not interpretations — stay neutral

Good outcome

Users complete key tasks without help

Their mental model matches yours

Feedback confirms core assumptions

Also a good outcome

Users struggle and you learn why

You discover you solved the wrong problem

You go back to Empathize — and that's fine

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PART 1 • WRAP-UP

The iterative loop — why it matters

"Design Thinking is not a straight line. Every test teaches you something new — and the best designers embrace that."

- ▶ Real-world design is **messy and non-linear** — that's normal and healthy
- ▶ Each loop through the phases gets you **closer to the right solution**
- ▶ Failing early with a paper prototype is infinitely cheaper than failing after launch
- ▶ Companies like Airbnb iterated their homepage **over 100 times** before finding what worked

MINDSET SHIFT

Stop thinking "did we succeed?" and start thinking "what did we learn?" Both outcomes move you forward.

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PART 2

02

AI as Your Design Partner

How Artificial Intelligence fits into — and accelerates — every phase of Design

Thinking.

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PART 2 · AI BASICS

What is AI? Plain English.



What AI is

Software that learns from data to recognise patterns, make predictions, and generate content.



How it learns

Trained on billions of examples (text, images, code). It predicts what comes next, not "understand".



Generative AI

Creates new content — text, images, code, audio. ChatGPT, Midjourney, GitHub Copilot.



What it's good at

Speed, scale, pattern recognition, language tasks, summarisation, generating options.



What it's bad at

True understanding, empathy, ethical reasoning, verifying facts, original thinking.



The right framing

Not a replacement for designers — a powerful assistant that amplifies your capabilities.

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PART 2 · AI BASICS

Generative AI — How does it actually work?

"Think of a Large Language Model (LLM) like ChatGPT as an incredibly well-read autocomplete — it predicts the most useful next word, millions of times per second."

- ▶ **Trained on trillions of words** from the internet, books, and code
- ▶ You give it a **prompt** — a question or instruction in plain language
- ▶ It generates a response that is **statistically likely to be useful** — not necessarily true
- ▶ Better prompts → better outputs. **Prompt engineering** is a real skill
- ▶ It has no memory between conversations unless you give it context

ANALOGY

AI is like a very fast, very well-read intern. Great at research, drafts, and volume — still needs your judgment, verification, and direction.

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PART 2 · AI AT EVERY PHASE

Phase 1 — Empathize with AI

Traditional empathy research takes weeks. AI compresses it — but never replaces real human conversations.

- ▶ **Interview synthesis** — paste transcripts, AI surfaces themes and patterns instantly
- ▶ **Sentiment analysis** — scan thousands of app reviews, social posts, or support tickets
- ▶ **Persona drafting** — AI proposes initial personas from your research for you to validate
- ▶ **Empathy map generation** — given research notes, AI structures a first-pass empathy map
- ▶ **Research question generation** — AI helps you design better interview questions

NON-NEGOTIABLE RULE

Never skip real conversations with users. AI can process data — it cannot feel what a user feels. Your empathy is still essential.

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Phase 2 — Define with AI

Crafting the right problem statement is hard. AI can challenge and enrich your thinking.

- ▶ **Problem reframing** — ask AI to rewrite your problem statement 10 different ways
- ▶ **"How Might We" generator** — turn insights into opportunity questions automatically
- ▶ **Assumption surfacing** — "What assumptions are hidden in this problem statement?"
- ▶ **Scope checker** — "Is this problem too broad, too narrow, or just right?"
- ▶ **Point of View templates** — AI fills in the structure, you validate with real data

PROMPT EXAMPLE

"Here are 5 insights from my user research: [paste]. Write 10 'How Might We' questions that address these themes."

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Phase 3 — Ideate with AI

AI can generate 100 ideas in 30 seconds — giving you raw material to spark your own creative thinking.

How to use AI in ideation

Generate 50+ initial ideas to break creative blocks

Combine unusual domains: "Spotify + hospital waiting rooms"

Reverse thinking: "What would make this problem worse?"

Analogous inspiration: "How do airlines handle this?"

Push beyond obvious: "Give me 5 counterintuitive ideas"

Tools

ChatGPT / Claude — text brainstorming

Miro AI — AI-assisted mind mapping

Notion AI — structured ideation docs

Stormz — AI-facilitated workshops

Mural AI — collaborative ideation boards

REMEMBER

AI gives you *quantity*. Your judgment provides *quality*. Use AI to expand, then use your brain to narrow.

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PART 2 · AI AT EVERY PHASE

Phase 4 — Prototype with AI

AI can turn a text description into a working UI mockup in seconds — compressing weeks of work into hours.



v0.dev

Type a description → get a working web UI component. Powered by Vercel.



Figma AI

Auto-layout, content generation, and design suggestions inside Figma.



Galileo AI

Describe your screen → AI generates full high-fidelity UI designs.



GitHub Copilot

AI writes the code for your prototype as you type — for interactive testing.



Midjourney / DALL-E

Generate concept art, mood boards, and visual directions instantly.



ElevenLabs

Generate realistic voice-overs for video or audio-based prototypes.

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PART 2 · AI AT EVERY PHASE

Phase 5 — Test with AI

AI assists in preparing, conducting, and analysing usability tests at scale.

- **Test script generation** — AI writes usability test tasks and follow-up questions
- **Simulated user feedback** — get an AI "first pass" review before real user testing
- **Session analysis** — tools auto-transcribe and summarise recorded user sessions

- **Pattern detection** — AI identifies recurring pain points across multiple test sessions
- **Heatmap interpretation** — Hotjar and similar tools use AI to explain click patterns

Tools

Maze — automated usability testing

UserTesting AI — session summaries

Hotjar Trends — behaviour analytics

Lookback — AI-coded observations

Still needs humans

Interpreting emotional context

Noticing unspoken hesitations

Building rapport with test participants

Making final design decisions

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PART 2 · THE IMPACT

Before vs After AI — what changes?

Phase	Without AI	With AI
Empathize	3 weeks to analyse 500 interviews	2–3 hours with AI synthesis
Define	Workshop of 6 people, half a day	Initial drafts in minutes, humans refine
Ideate	Brainstorm: 40 ideas in 1 hour	AI generates 200 ideas in 2 minutes
Prototype	Developer needed, 1–2 weeks	Working UI from text prompt in minutes
Test	Manual review of session recordings	Auto-transcription + AI-coded themes

NET RESULT

Design cycles that took months can now take weeks. More iterations → better products → happier users. AI doesn't replace the process — it *accelerates* it.

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PART 2 · THE HUMAN ROLE

What only humans can do

As AI handles more tasks, these distinctly human capabilities become more valuable — not less.



Empathy

Genuinely feeling a user's frustration or joy. AI can describe emotions — not experience them.



Ethical judgment

Deciding what is right, fair, and inclusive. AI reflects its training biases uncritically.



Cultural context

Understanding the unspoken rules of a community, culture, or specific moment in history.



Creative vision

The spark that connects unrelated ideas into something genuinely new and meaningful.



Facilitation

Leading workshops, building trust, reading body language, navigating group dynamics.



Critical thinking

Questioning AI output, spotting errors, and deciding when NOT to follow the suggestion.

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PART 3

03

Practice, Risks & Your Future

Real-world cases, responsible design, your toolkit, and what this all means for your career.

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Real World — A win with AI

"We needed to understand pain points across 500 patient interviews from three hospitals. Manually, that's three weeks of analysis. With AI, we had clustered themes in two hours."

— Health-tech design team lead

- ▶ AI surfaced a recurring theme the team had completely missed in manual reviews
- ▶ Theme: patients feared calling to reschedule because they felt judged
- ▶ Design response: added a no-explanation-needed reschedule button
- ▶ **Result: 30% reduction in appointment no-shows within 3 months**

WHY IT WORKED

The team used AI to process data faster — but they still spoke to real patients to validate the insight before building anything.

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Real World — A cautionary tale

"We used AI to generate all our user personas. They were detailed, plausible, and completely fictional. We built for people who don't exist."

— Startup founder, post-mortem interview

- ▶ Team skipped real user interviews — "AI can do that"
- ▶ AI generated convincing personas based on generic internet data
- ▶ Product launched, real users didn't recognise themselves in it
- ▶ **Result: 6 months and \$200K wasted. Product shut down.**

THE LESSON

AI is a brilliant accelerator for processing data you already have. It is a terrible substitute for data you haven't collected yet. *You still have to talk to people.*

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PART 3 · RESPONSIBLE DESIGN

Risk 1 — Bias & Hallucinations

Bias in AI outputs

AI is trained on historical data — which reflects historical biases

Older data may underrepresent women, minorities, non-Western users

Biased AI → biased insights → products that exclude people

Always ask: "Who is missing from this data?"

Hallucinations

AI confidently states false information — no warning label attached

It cannot distinguish fact from plausible-sounding fiction

Never cite AI outputs as facts without verifying from primary sources

Especially dangerous in medical, legal, and financial design contexts

RULE

Treat all AI-generated facts like a first draft from a new intern — promising, but always verify before acting on them.

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PART 3 · RESPONSIBLE DESIGN

Risk 2 — Privacy & Over-reliance

Privacy risks

Never paste real user data (names, emails, health info) into public AI tools

Data entered into ChatGPT (free tier) may be used for training

Use anonymised or synthetic data for AI-assisted analysis

Check your company/institution's AI usage policy

Over-reliance

Designers who stop interviewing users lose the skill

Accepting AI output without critique creates "lazy design"

AI approval is not a substitute for human validation

Critical thinking is a muscle — don't let it atrophy

GOLDEN RULE

Use AI to go *faster* — never to *skip thinking*. The moment you stop questioning AI

PART 3 • RESPONSIBLE DESIGN

Responsible AI Design — Checklist

Before shipping anything built with AI assistance, ask yourself:

- ✓ Have I spoken to real users — not just relied on AI-generated insights?
- ✓ Have I verified AI-generated facts from primary sources?
- ✓ Does my design work for users across different backgrounds and abilities?
- ✓ Have I checked whether AI outputs reflect biases in the training data?
- ✓ Did I anonymise user data before using any AI tool?
- ✓ Have I been transparent with users where AI is used in the product?
- ✓ Am I prepared to explain every AI-assisted decision I made?
- ✓ Is there a human in the loop for consequential decisions?

PART 3 • YOUR TOOLKIT

Practical AI Tools for Each Phase

Phase	Tool(s)	What it does	Free?
Empathize	ChatGPT, Claude	Synthesise interviews, find themes	Yes
Define	ChatGPT, Notion AI	Reframe problems, generate HMW questions	Yes
Ideate	Miro AI, Mural AI	AI-assisted brainstorming & mind maps	Limited
Prototype	v0.dev, Figma AI, Galileo	Text-to-UI, design generation	Yes / Limited
Test	Maze, Hotjar, Lookback	Automated usability + session analysis	Limited

Phase	Tool(s)	What it does	Free?
Visuals	Midjourney, DALL-E	Mood boards, concept art	Limited
Code	GitHub Copilot	AI-assisted coding for prototypes	Free for students

START HERE TODAY

Open ChatGPT (free), paste in a problem statement, and ask it to generate 20 "How Might We" questions. That's Design Thinking + AI in under 5 minutes.

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KEY TAKEAWAYS

What to remember from today

- **Design Thinking** = empathy-first problem solving in 5 iterative phases
- **Most solutions fail** not from bad tech, but from misunderstanding the problem
- **AI accelerates every phase** — from synthesis to prototyping to testing
- **Human empathy, judgment, and ethics** are the irreplaceable layers on top
- **Beware** bias, hallucinations, privacy risks, and skipping real user research
- **Use AI to go faster** — never to skip thinking
- **Start today** — the tools are free, and the skills compound

Thank you. Questions? Discussion? Let's talk.

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